

Q1. Google PageRank - Ranking billions of webpages using eigenvectors.

→ The web can be represented as a directed graph, where each webpage is a node and each hyperlink represents a connection between two pages. A transition matrix is constructed in which each entry represents the probability of moving from one page to another. PageRank then uses dominant eigenvector of this matrix to determine the relative importance of pages.

$$P = \begin{bmatrix} 0 & 0.5 & 1 \\ 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$

columns represent current page and rows represent the page being visited next.

The pagerank vector x , containing the A, B, C importance, is obtained from

$$Px = \lambda x$$

Characteristic equation,

$$\begin{aligned} & -\lambda \times 0 \rightarrow \\ & -\lambda(\lambda^2) - 0.5(-0.5\lambda) + (0.25 + 0.5\lambda) = 0 \\ \Rightarrow & -\lambda^3 + 0.75\lambda + 0.25 = 0 \end{aligned}$$

solving, $\lambda_{max} = 1$

corresponding $x = \begin{bmatrix} 1 \\ 0.5 \\ 0.75 \end{bmatrix}$

This gives the relative ranking of the three pages. Here $A > C > B$.

Rough

$$\begin{aligned} -x + y/2 + z &= 0 \\ x/2 - y &= 0 \\ x/2 + y/2 - z &= 0 \\ x &= 1 \\ y &= 0.5 \\ z &= 0.75 \end{aligned}$$

Q2. GPS location Tracking

The distance from an unknown GPS position (x, y) to a known ~~location~~ reference point (a, b) is given by,

$$(x-a)^2 + (y-b)^2 = d^2$$

With measurements from multiple satellites, we get multiple quadratic equations. By subtracting one equation from another, the ' $x^2 + y^2$ ' terms cancel, leaving linear equations in ~~3 as~~ x and y (2D). These can be written as a matrix equation $Ax = b$.

Suppose, the unknown position is (x, y) and distances from three known points are:

$$x^2 + y^2 = 25 \quad \text{--- (1)}$$

$$(x-4)^2 + y^2 = 9 \quad \text{--- (2)}$$

$$x^2 + (y-4)^2 = 5 \quad \text{--- (3)}$$

Using first equation,

$$\textcircled{2} \rightarrow 8x = 32$$

$$\textcircled{3} \rightarrow 8y = 36$$

In $Ax = b$ form,

$$\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 32 \\ 36 \end{bmatrix}$$

$$\text{solving, } \begin{aligned} x &= 4 \\ y &= 4.5 \end{aligned}$$

Q3. Medical Imaging (CT/MRI Reconstruction)

→ Medical scanners collect measurements from different directions or locations. These measurements are represented mathematically as equations involving the unknown image values. Matrix methods are then used to reconstruct the original image.

Suppose an image has two unknown pixel values, x and y , and the scanner produces,

$$x + y = 5$$

$$x - y = -1$$

In $Ax = b$ form,

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

Solving $x = 3$, $y = 2$

This is how pixel values can be reconstructed from the measured data.

Q4. Structural Engineering Vibrations

→ Building and bridges can be modeled using mass and stiffness matrices. The natural frequencies are obtained by solving the eigenvalues of K/M [K = stiffness matrix, M = mass matrix]. The eigenvalues determine the frequencies and the eigenvectors describe the corresponding vibration mode.

Let's take the mass matrix (M) to be the identity matrix to simplify the problem.

So, $Kx = \lambda Mx$ become,

$$Kx = \lambda x$$

Suppose $K = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$

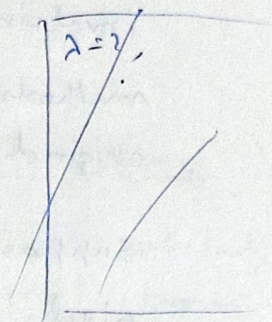
characteristic eqⁿ,

$$(2-\lambda)(8-\lambda) = 0$$

$$\Rightarrow \lambda = 2, 8$$

The natural frequencies become $\sqrt{2}$ and $\sqrt{8}$.

The modes, correspondingly are, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ for both.



5. Wireless Communication (MIMO Systems)

MIMO uses multiple transmitting and receiving antennas. The received signals are related to the transmitted signals through a channel matrix. Matrix methods such as eigenvalue decomposition or SVD can separate independent signal components.

Suppose $y = Hx$, $H = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

if x , the transmitted signal is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

y , the received signal become,

$$H \cdot x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

And the two signals are transmitted independently through two different channels since the H matrix is diagonal.

Q6. Economics and Market Equilibrium

Market equilibrium occurs when quantity supplied equals quantity demanded. Supply and demand equations can be written as simultaneous equations and solved using matrix equations.

Say, x = equilibrium quantity of one economic variable, i.e. demand

y = equilibrium value of another economic variable, i.e. supply.

A is a coefficient matrix describing the relationship between them.

v is a given vector & relation between vectors.

We take $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$

$$\therefore \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$$

solving $x = 4$, $y = 6$

$\therefore$ The equilibrium price is 4 units,
and the equilibrium quantity is 6 units

Q7. Computer Graphics (3D Transformations)

3D objects are represented using coordinate vectors. Transformations such as rotation and scaling are performed by multiplying suitable matrices with them.

Consider a home with coordinates,

$$A = \begin{bmatrix} 1 & 3 & 3 & 1 & 2 \\ 1 & 1 & 3 & 3 & 5 \end{bmatrix}$$

90° rotation matrix is,

$$R_{90} = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$\therefore$ Rotated $A' \equiv R_{90} \cdot A = \begin{bmatrix} -1 & -1 & -3 & -3 & -5 \\ 1 & 3 & 3 & 1 & 2 \end{bmatrix}$

Scaling matrix $S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ [scaling both axes with $k=2$]

$\therefore$ Scaled $A'' = SA = \begin{bmatrix} 2 & 6 & 6 & 2 & 4 \\ 2 & 2 & 6 & 2 & 10 \end{bmatrix}$

Q8. Cryptography and Coding Theory - Encoding/Decoding

A message is a datapoint in n -dimensional vector space. A key matrix is multiplied with these vectors to produce encoded data. The inverse of the same matrix is used for decoding the message back.

Let the message vector be $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

and the encoding matrix be $K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Then - encoded message $= Kx = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Decoded message $= K^{-1}Kx = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix} (x)$$

Q9. Epidemiology - Disease spread

The population can be divided into groups such as susceptible and infected individuals. A transition matrix describes how people move between these groups from one time period to the next.

Suppose $x_0 = \begin{bmatrix} 90 \\ 10 \end{bmatrix}$

represents 90 susceptible and 10 infected people. Let, $T = \begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$

The transition matrix basically predicts the population distribution in the next time period.

$$x_1 = Tx_0 = \begin{bmatrix} 82 \\ 18 \end{bmatrix}$$

is the x matrix in next iteration (time period)

Q10. Air traffic control - Aircraft positioning

Aircraft position can be determined using measurements such as distances or coordinates from multiple reference points. These measurements produce simultaneous equations, which can be solved using matrix methods to determine unknown position.

Suppose, an aircraft position satisfies

$$x + y = 10 \quad - \textcircled{1}$$

$$x - y = 2 \quad - \textcircled{2}$$

Solving gives, $x = 6$, $y = 4$,

$\therefore$ The aircraft position is $(6, 4)$

Q11.

$$M_1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$M_1^T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$\begin{aligned} M_1^T M_1 &= \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \end{aligned}$$

Let eigenvalue be λ ,

$$\det \begin{pmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{pmatrix} = 0$$

$$\therefore (2-\lambda)^2 - 2 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\therefore \lambda = +1, +3$$

$\therefore$ singular value, $\sigma = \sqrt{\lambda} = \sqrt{1}, \sqrt{3}$

$$= 1, 1.7321$$

eigenvalue for $\lambda = 3$,

$$-x - y = 0 \quad \text{--- i)}$$

$$-x - y = 0 \quad \text{--- ii)}$$

$$\therefore y = -x$$

$$\therefore \text{Eigenvector } v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{normalizing, } v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$$

For $\lambda = 1$,

$$x - y = 0 \Rightarrow x = y$$

$$-x + y = 0$$

$$\therefore v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \text{ normalizing, } v = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$$

$$\therefore V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}; \Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}$$

$$U_1 = \frac{M_1 v_1}{\sigma_1} = \frac{\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0.2071 \\ -0.7071 \end{bmatrix}}{1.7321}$$

$$= \begin{bmatrix} 0.8165 \\ -0.4082 \\ 0.4082 \end{bmatrix}$$

$$U_2 = \frac{M_1 v_2}{\sigma_2} = \frac{\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}}{1} = \begin{bmatrix} 0 \\ 0.7071 \\ 0.7071 \end{bmatrix}$$

$$\therefore \boxed{U} = \begin{bmatrix} 0.8165 & 0.0 \\ -0.4082 & 0.7071 \\ 0.4082 & 0.7071 \end{bmatrix}$$

$$\boxed{\Sigma} = \begin{bmatrix} 1.7321 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{V^T} = \begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & -2 \\ 1 & 2 & 0 \end{bmatrix}$$

$$m_2^T = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix}$$

$$m_2^T m = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 5 & 1 \\ 3 & 1 & 5 \end{bmatrix}$$

Find $\det (M_2^T M - \lambda I) = 0$

$$\begin{vmatrix} 3-\lambda & 3 & 3 \\ 3 & 5-\lambda & 1 \\ 3 & 1 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda^3 + 13\lambda^2 - 28\lambda = 0$$

$$\Rightarrow -\lambda(\lambda-4)(\lambda-9) = 0$$

$$\therefore \lambda = 0, 4, 9$$

$$\therefore \sigma = 0, 2, 3$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \text{ normalised } v_1 = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \text{ normalised } v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}, \text{ normalised } v_3 = \frac{1}{\sqrt{14}} \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$$

$$\therefore \boxed{V} = \begin{bmatrix} 1/3 & 1/\sqrt{2} & -2/\sqrt{14} \\ 2/3 & 0 & 3/\sqrt{14} \\ 2/3 & -1/\sqrt{2} & -1/\sqrt{14} \end{bmatrix}$$

$$\boxed{\Sigma} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$u_i = \frac{1}{\sigma_i} M_2 v_i$$

$$u_1 = \frac{1}{3} \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{1}{\sqrt{14}} \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1/3 & 0 & 4/\sqrt{14} \\ -1 & 1/\sqrt{2} & 1/\sqrt{14} \\ 5/3 & 1/\sqrt{2} & -1/\sqrt{14} \end{bmatrix}$$

Q12. Netflix recommendation

A recommendation system such as Netflix uses user-item rating matrix to determine preference.

Suppose there are 3 users and 4 movies.

$$R = \begin{bmatrix} 5 & 4 & 0 & 1 \\ 3 & 0 & 2 & 1 \\ 1 & 2 & 5 & 4 \end{bmatrix}$$

Row represent users
column represent movies

The matrix can be factored by $R \approx UV^T$

U represents the users in a smaller latent feature space,

V represents the movies in the latent feature space.

$$U = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 3 \\ 2 & 1 \end{bmatrix}$$

decomposing

$$U = \begin{bmatrix} -0.6179 & -0.6675 & 0.4153 \\ -0.4536 & -0.1285 & -0.8818 \\ -0.6420 & 0.7333 & 0.2239 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 8.66 & 0 & 0 \\ 0 & 5.23 & 0 \\ 0 & 0 & 2.55 \end{bmatrix}$$

$$V^T = \begin{bmatrix} -0.64 & -0.43 & -0.47 & -0.42 \\ -0.59 & -0.23 & 0.65 & 0.41 \\ -0.47 & 0.82 & -0.25 & 0.17 \end{bmatrix}$$

Predicted matrix :

$$\begin{bmatrix} 5 & 4 & 0 & 1 \\ 4 & 0 & 2 & 1 \\ 1 & 2 & 5 & 4 \end{bmatrix}$$

So, factorization allow recommender system to discover hidden relationships between users and items and predict a user's preference for an unrated item.

Q13. Machine learning - Feature spaces

Each datapoint is a vector in feature space.

Suppose we have the three observations described by two features.

$$X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \end{bmatrix}$$

The standard basis of this space is

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Instead of representing the data in original feature direction we can denote a new basis containing directions that capture the most variation in data.

This is the basic idea behind PCA.

If the new basis vectors are p_1 and p_2 , the same data can be represented using coordinates along these new direction.

The first principal component is chosen in the direction of maximum variance, while subsequent components capture the remaining variances (sorted).

We can stop this after a certain percentage of variation (say 90%, or 95% or 99%) is captured by the new axes.